Logarithm: Questions and Full Solutions

Question 01

English

Let S be the set of all real numbers x satisfying

$$\log_{x^2+4x+2}(4x+6) = \log_{4x+6}\left(x^2+4x+2\right).$$

Then the sum of all the elements of S is equal to

(A) $-\dfrac{11}{2}$
(B) $-\dfrac{1}{2}$
(C) $\dfrac{3}{2}$
(D) 2

Only one option is correct.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं x का समुच्चय है जो समीकरण

$$\log_{x^2+4x+2}(4x+6) = \log_{4x+6}\left(x^2+4x+2\right)$$

को संतुष्ट करती हैं। तब S के सभी अवयवों का योग बराबर है

(A) $-\dfrac{11}{2}$
(B) $-\dfrac{1}{2}$
(C) $\dfrac{3}{2}$
(D) 2

केवल एक विकल्प सही है।

Solution

Write $A = x^2+4x+2$ and $B = 4x+6$. Both logarithms are defined only when $A > 0,\ A \neq 1,\ B > 0,\ B \neq 1$.

$B > 0 \Rightarrow x > -\dfrac{3}{2}$. $A > 0 \Rightarrow x < -2-\sqrt{2}$ or $x > -2+\sqrt{2}$; with $x > -\dfrac{3}{2}$ this leaves $x > \sqrt{2}-2$.

$A = 1 \Rightarrow x^2+4x+1 = 0 \Rightarrow x = -2 \pm \sqrt{3}$; only $x = \sqrt{3}-2$ lies in $x > \sqrt{2}-2$. $B = 1 \Rightarrow x = -\dfrac{5}{4} < \sqrt{2}-2$.

So the domain of the equation is $D = \left(\sqrt{2}-2, \infty\right) \setminus \left\{\sqrt{3}-2\right\}$.

Put $u = \log_A B$. By the change of base, $\log_B A = \dfrac{1}{u}$ ($u \neq 0$ since $B \neq 1$). The equation is $u = \dfrac{1}{u} \Rightarrow u^2 = 1 \Rightarrow u = 1$ or $u = -1$.

Case $u = 1$: $B = A \Rightarrow x^2+4x+2 = 4x+6 \Rightarrow x^2 = 4 \Rightarrow x = 2$ or $x = -2$.

$x = -2$: $A = B = -2 < 0$, not in D. $x = 2$: $A = B = 14$, in D; both sides equal 1.

Case $u = -1$: $B = \dfrac{1}{A} \Rightarrow (x^2+4x+2)(4x+6) = 1 \Rightarrow 4x^3+22x^2+32x+11 = 0$.

$x = -\frac{1}{2}$ is a root: $-\frac{1}{2}+\frac{11}{2}-16+11 = 0$. Dividing, $4x^3+22x^2+32x+11 = (2x+1)(2x^2+10x+11)$.

$2x^2+10x+11 = 0 \Rightarrow x = \frac{-5 \pm \sqrt{3}}{2} \approx -1.63,\ -3.37$. Both are less than $-\frac{3}{2}$, so $B < 0$ (here $A < 0$ too, which is why $AB = 1$ holds); both rejected.

$x = -\frac{1}{2}$: $A = \frac{1}{4} - 2 + 2 = \frac{1}{4}$, $B = 4$; in D, and $\log_{1/4} 4 = -1 = \log_4 \frac{1}{4}$. Accepted.

Hence $S = \left\{2, -\frac{1}{2}\right\}$ and the sum of its elements is $2 - \frac{1}{2} = \frac{3}{2}$.

(Ignoring the domain gives all five candidates with sum $-\frac{11}{2}$; keeping $x = -2$ gives $-\frac{1}{2}$.)

Correct Option: (C)

Question 02

English

For $x \in \left(0, \frac{1}{10}\right)$, let

$$P(x) = \log_{1+x}(1+2x) \cdot \log_{1+2x}(1+3x) \cdot \log_{1+3x}(1+4x) \cdots \log_{1+9x}(1+10x),$$

the product of the nine factors $\log_{1+kx}(1+(k+1)x)$, $k = 1, 2, \ldots, 9$. Then the value of $\lim\limits_{x \to 0^+} \frac{10 - P(x)}{x}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

$x \in \left(0, \frac{1}{10}\right)$ के लिए माना कि

$$P(x) = \log_{1+x}(1+2x) \cdot \log_{1+2x}(1+3x) \cdot \log_{1+3x}(1+4x) \cdots \log_{1+9x}(1+10x),$$

जो $k = 1, 2, \ldots, 9$ के लिए नौ गुणनखंडों $\log_{1+kx}(1+(k+1)x)$ का गुणनफल है। तब $\lim\limits_{x \to 0^+} \frac{10 - P(x)}{x}$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Each factor is defined for $x \in \left(0, \frac{1}{10}\right)$: every base $1 + kx > 1$ and every argument is positive.

By the change of base, $\log_{1+kx}(1+(k+1)x) = \frac{\ln(1+(k+1)x)}{\ln(1+kx)}$.

In the product over $k = 1, \ldots, 9$ the numerator of each factor cancels the denominator of the next, leaving $P(x) = \frac{\ln(1+10x)}{\ln(1+x)}$.

So $P(x) \to 10$ as $x \to 0^+$ and the limit is of the form $\frac{0}{0}$.

Use $\ln(1+u) = u - \frac{u^2}{2} + O(u^3)$: $\ln(1+10x) = 10x - 50x^2 + O(x^3)$ and $\ln(1+x) = x - \frac{x^2}{2} + O(x^3)$.

$P(x) = \frac{10 - 50x + O(x^2)}{1 - \frac{x}{2} + O(x^2)} = (10 - 50x)\left(1 + \frac{x}{2}\right) + O(x^2) = 10 - 45x + O(x^2)$.

Hence $\frac{10 - P(x)}{x} = 45 + O(x) \to 45$.

(In general the product up to base $1+(n-1)x$ gives $\dfrac{n(n-1)}{2}$; here $n=10$.)

Final Answer: 45

Question 03

English

Let $f:(0,\infty)\setminus\left\{\dfrac{1}{2},8\right\}\to\mathbb{R}$ be defined by

$$f(x)=\log_{x/8}(4x)\cdot\log_{2x}4.$$

The range of f is $(-\infty,\alpha]\cup[\beta,\infty)$, where $\alpha<\beta$. Then the value of $\alpha+\beta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f:(0,\infty)\setminus\left\{\dfrac{1}{2},8\right\}\to\mathbb{R}$

$$f(x)=\log_{x/8}(4x)\cdot\log_{2x}4$$

द्वारा परिभाषित है। f का परिसर $(-\infty,\alpha]\cup[\beta,\infty)$ है, जहाँ $\alpha<\beta$ है। तब $\alpha+\beta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The bases $\dfrac{x}{8}$ and $2x$ are positive and differ from 1 exactly when $x>0,\ x\neq 8,\ x\neq\dfrac{1}{2}$, which is the given domain.

Let $t=\log_2 x$; as x runs over the domain, t runs over $\mathbb{R}\setminus\{3,-1\}$.

By the product, quotient and power rules and the change of base: $\log_{x/8}(4x)=\dfrac{\log_2 4+\log_2 x}{\log_2 x-\log_2 8}=\dfrac{t+2}{t-3}$ and $\log_{2x}4=\dfrac{2}{1+t}$.

So $f=\dfrac{2(t+2)}{(t-3)(t+1)}$. A value y is in the range iff $2(t+2)=y(t^2-2t-3)$ has a root $t\neq 3,-1$, i.e. iff $yt^2-(2y+2)t-(3y+4)=0$ has such a root.

At $t=3$ the left side is $9y-6y-6-3y-4=-10\neq 0$; at $t=-1$ it is $y+2y+2-3y-4=-2\neq 0$. So the excluded values of t are never roots and remove nothing.

$y=0$: the equation is $-2t-4=0$, $t=-2$, allowed; so 0 is in the range.

$y\neq 0$: a real root exists iff the discriminant $(2y+2)^2+4y(3y+4)=16y^2+24y+4\geq 0$, i.e. $4y^2+6y+1\geq 0$.

The roots of $4y^2+6y+1=0$ are $\dfrac{-3\pm\sqrt{5}}{4}$, so the range is $\left(-\infty,\dfrac{-3-\sqrt{5}}{4}\right]\cup\left[\dfrac{-3+\sqrt{5}}{4},\infty\right)$ (it contains 0, as it must).

Thus $\alpha=\dfrac{-3-\sqrt{5}}{4}$, $\beta=\dfrac{-3+\sqrt{5}}{4}$ and $\alpha+\beta=-\dfrac{3}{2}=-1.50$.

Final Answer: -1.50

Question 04

English

Let S be the set of all real numbers $x > 0$, $x \neq 1$, for which the infinite series

$$1 - \log_x 4 + (\log_x 4)^2 - (\log_x 4)^3 + \cdots$$

is convergent and its sum is equal to $\log_{1/2}(4x)$. Then the value of $\sum_{x \in S} \log_2 x$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं $x > 0$, $x \neq 1$ का समुच्चय है जिनके लिए अनंत श्रेणी

$$1 - \log_x 4 + (\log_x 4)^2 - (\log_x 4)^3 + \cdots$$

अभिसारी है तथा इसका योग $\log_{1/2}(4x)$ के बराबर है। तब $\sum_{x \in S} \log_2 x$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $t = \log_2 x$; $x > 0,\ x \neq 1$ means $t \in \mathbb{R} \setminus \{0\}$.

By the change of base, $\log_x 4 = \dfrac{\log_2 4}{\log_2 x} = \dfrac{2}{t}$.

The series $1 - \dfrac{2}{t} + \left(\dfrac{2}{t}\right)^2 - \cdots$ is a geometric series with first term 1 and common ratio $r = -\dfrac{2}{t}$.

An infinite GP converges iff $|r| < 1$, i.e. $\left|\dfrac{2}{t}\right| < 1 \iff |t| > 2$; its sum is then $\dfrac{1}{1-r} = \dfrac{1}{1+\frac{2}{t}} = \dfrac{t}{t+2}$.

By the power and product rules, $\log_{1/2}(4x) = -\log_2(4x) = -(2+t)$.

Equating: $t = -(t+2)^2 \Rightarrow t^2 + 5t + 4 = 0 \Rightarrow (t+1)(t+4) = 0 \Rightarrow t = -1$ or $t = -4$.

$t = -1$ ($x = \dfrac{1}{2}$): $|t| = 1 < 2$; the ratio is $r = 2$ and the series $1 + 2 + 4 + \cdots$ diverges. The formal value $\dfrac{t}{t+2} = -1$ equals $\log_{1/2} 2 = -1$, but this is not a sum; rejected.

$t = -4$ ($x = \dfrac{1}{16}$): $|t| = 4 > 2$, $r = \dfrac{1}{2}$, sum $= \dfrac{1}{1-\frac{1}{2}} = 2$, and $\log_{1/2} \dfrac{1}{4} = 2$. Accepted.

So $S = \left\{\dfrac{1}{16}\right\}$ and $\sum_{x \in S} \log_2 x = -4$ (keeping the divergent case would give -5).

Final Answer: -4

Question 05

English

Let S be the set of all real numbers $x > 0$, $x \neq \dfrac{1}{2}$, $x \neq 2$, satisfying

$$\log_{2x}(16x) \cdot \log_{x/2}\left(\frac{x}{16}\right) = -4.$$

Then the sum of all the elements of S is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S उन सभी वास्तविक संख्याओं $x > 0,\ x \neq \dfrac{1}{2},\ x \neq 2$ का समुच्चय है जो

$$\log_{2x}(16x) \cdot \log_{x/2}\left(\frac{x}{16}\right) = -4$$

को संतुष्ट करती हैं। तब S के सभी अवयवों का योग _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The bases $2x$ and $\dfrac{x}{2}$ are positive and different from 1 for $x > 0,\ x \neq \dfrac{1}{2},\ x \neq 2$; the arguments $16x$ and $\dfrac{x}{16}$ are positive.

Let $t = \log_2 x$, so $t \neq -1$ and $t \neq 1$.

Change of base to base 2: $\log_{2x}(16x) = \dfrac{\log_2 16 + \log_2 x}{\log_2 2 + \log_2 x} = \dfrac{t+4}{t+1}$ and $\log_{x/2}\dfrac{x}{16} = \dfrac{t-4}{t-1}$.

The equation becomes $\dfrac{(t+4)(t-4)}{(t+1)(t-1)} = -4 \Rightarrow t^2 - 16 = -4(t^2-1) \Rightarrow 5t^2 = 20 \Rightarrow t = \pm 2$.

(The left side is unchanged when t is replaced by $-t$, i.e. x by $\dfrac{1}{x}$, so the roots come as a reciprocal pair.)

$t = 2 \Rightarrow x = 4$: $\log_8 64 \cdot \log_2 \dfrac{1}{4} = 2 \cdot (-2) = -4$. $t = -2 \Rightarrow x = \dfrac{1}{4}$: $\log_{1/2} 4 \cdot \log_{1/8} \dfrac{1}{64} = (-2) \cdot 2 = -4$. Both valid.

So $S = \left\{4, \dfrac{1}{4}\right\}$ and the sum of its elements is 4.25.

Final Answer: 4.25

Question 06

English

Let P be the product of the first 10 terms of the geometric progression $6,\ 3,\ \dfrac{3}{2},\ \dfrac{3}{4},\ \ldots$. It is given that $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$. Then the mantissa of $\log_{10} P$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि P गुणोत्तर श्रेढ़ी $6,\ 3,\ \dfrac{3}{2},\ \dfrac{3}{4},\ \ldots$ के प्रथम 10 पदों का गुणनफल है। दिया है कि $\log_{10} 2 = 0.3010$ तथा $\log_{10} 3 = 0.4771$ है। तब $\log_{10} P$ का अपूर्णांश (mantissa) _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The progression has first term $a = 6$ and common ratio $r = \dfrac{1}{2}$, so the k-th term is $6\left(\dfrac{1}{2}\right)^{k-1}$.

Product of the first 10 terms: $P = \prod_{k=1}^{10} 6\left(\dfrac{1}{2}\right)^{k-1} = 6^{10}\left(\dfrac{1}{2}\right)^{0+1+\cdots+9} = 6^{10}\left(\dfrac{1}{2}\right)^{45}$.

Since $6^{10} = 2^{10}3^{10}$, $P = \dfrac{3^{10}}{2^{35}}$.

$\log_{10} P = 10\log_{10} 3 - 35\log_{10} 2 = 4.7710 - 10.5350 = -5.7640$.

The characteristic is the greatest integer not exceeding $\log_{10} P$, and the mantissa is the non-negative

part: $-5.7640 = -6 + 0.2360$.

So the characteristic is -6 and the mantissa is 0.2360, which is 0.24 to two decimal places (the exact value $0.2352\ldots$ gives the same rounding).

(Taking 0.764 as the mantissa is the usual mistake.)

Final Answer: 0.24